COLLEGE MANAGEMENT SYSTEM

[Project code: PRJ 251]

Submitted to:

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A SECOND YEAR PROJECT REPORT

In partial fulfillment of the requirement for the degree of

BACHELOR OF COMPUTER APPLICATION(BCA)

April 2024

CERTIFICATE FROM THE EXTERNAL EXAMINER

This is to certify the work carried out by "Ammar Bahadur Kunwar, Amrit Shrestha, Keshab Chidi, Nabin Chaudhary and Sudip Rana" for the completion of the project entitled "COLLEGE MANAGEMENT SYSTEM" as the fulfillment of the requirement of the Degree of Bachelor of Computer Application has been accomplished, presented, and demonstrated successfully.

During presentation, I have found the students to be enthusiastic, hardworking, punctual, and ready with skills to undertake any work related to IT and Computer Application.

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BONAFIDE-CERTIFICATE

Certified that this project "**College Management System**" is the bonafide work of "AMMAR BAHADUR KUNWAR, AMRIT SHRESTHA, KESHAB CHIDI, NABIN CHAUDHARY and SUDIP RANA" who carried out the project or under my supervision.

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Mr. Abdul Haq	**Mr. Suraj Pandey**
IT coordinator	**Project Supervisor**
Department of IT	**Department of IT**

DECLARATION

The project documentation in this report was carried out by five below mentioned 4th semester undergraduate students of Bachelor of Computer Application (BCA) for the fulfillment of the requirement for the project of Four semester under the supervision of **Mr. Suraj Pandey.**

Any kind of reproduction of the project and its part for commercial purpose is strictly prohibited without prior written permission of the author and university. However, this report is open and free for any kind of academic purpose.

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Project Supervisor

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PREFACE

In today's dynamic educational landscape, efficient management of quizzes is essential for effective learning outcomes. With advancing technology, traditional quiz tracking methods are being replaced by automated systems that promise enhanced accuracy, convenience, and transparency.

This college project delves into the development of a **"COLLEGE MANAGEMENT SYSTEM"**, aimed at streamlining the process of conducting quizzes while providing valuable insights for educators. It explores the conceptualization, design, and implementation of this system, offering a comprehensive understanding of its functionalities, benefits, and potential challenges.

ACKNOWLEDGEMENT

The task of developing any project without any previous experiences or any detail knowledge of any programming language is very difficult task. We sincerely express our gratitude and sincere respect to all of them who directly and indirectly supported and helped us to make our project successfully.

We sincerely thank CRIMSON COLLEGE OF TECHNOLOGY for providing us with this opportunity to improve our knowledge and support in completion of our project. We thank our principal, DEVI PRASAD BHANDARI for providing us the necessary infrastructure. We are also thankful to Mr. Suraj Pandey, project supervisor who cooperated and provided information about the project and without whose supervision or would not have been possible to get up to this level of project. We thank our IT coordinator, Mr. Abdul Hak for his support and motivation. We thank all the teaching and non-teaching staff members of the Information Science Department for providing the necessary guidance, help and cooperation. Likewise, we are very much grateful to all our senior friends who provided the format of already prepared project as a reference and shared the experience of this project developing.

Although our best attempt to develop the project error free and effective, there may be error in this project. We don't say that we have prepared the best one. Any errors, omissions and valuable suggestion regarding to this project from anyone will be thankfully acknowledged.

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ABSTRACT

In response to the growing demand for dynamic and interactive assessment tools in the realm of online education, this research presents a sophisticated Quiz System designed to elevate the learning experience. With a user-centric approach, the system caters to both administrators and participants, offering a robust platform for quiz creation, management, and engagement.

For users, the Quiz System provides an intuitive interface facilitating seamless access to quizzes, answering questions, and reviewing results. Real-time feedback mechanisms and performance analytics contribute to a personalized learning journey, fostering a deeper understanding of the subject matter. Administrators benefit from powerful tools enabling the creation of diverse quizzes, real-time monitoring of participant progress, and comprehensive analysis of quiz outcomes.

The architecture of the Quiz System integrates cutting-edge web development technologies and efficient database management, ensuring scalability, security, and optimal performance. The user-friendly design emphasizes accessibility, while the system's robust functionality aims to create a dynamic and engaging learning environment.

This research delves into the development, implementation, and potential impact of the Quiz System on educational practices. By blending technological innovation with pedagogical principles, the system addresses the evolving needs of educators and learners. The study contributes valuable insights to the broader discourse on effective e-learning tools, offering strategies for educational advancement in the digital age.

In conclusion, the Quiz System represents a significant step towards enhancing the online learning experience, promising to reshape the landscape of educational assessment tools. The findings of this research position the system as a valuable asset in the ongoing evolution of e-learning methodologies, providing a foundation for further exploration and refinement.

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UNIT 1: INTRODUCTION OF THE PROJECT

1.1 Introduction of Project:

The College Management System, a comprehensive software solution crafted with VB.NET, aims to serve to the intricate needs of academic institutions. This system is carefully designed to streamline and automate diverse administrative processes within a college environment. By doing so, it endeavors to elevate efficiency levels while providing a user-friendly platform for the seamless management of details on students, and their fees.

"College Management System" is developed as per seeing the increasing requirement to speed up the work and incorporate a new work culture. Thus, a new software has been proposed to reduce manual work, improving work efficiency, and saving time.

1.2 Problem Statement:

After examining the manual administrative processes in educational institutions, such as enrollment and fee payments, we've identified significant challenges. These include:

- Time-consuming tasks and error-prone processes due to the large records of students.
- Data fragmentation across different systems and paper-based records, leading to inefficiencies in information management.
- Limited scalability and flexibility of records.
- Lack of integrated system hampers decision-making, and user experience.

1.3 Objectives:

The primary objectives of the College Management System are as follows:

- Efficient Data Management:
  The College Management System ensures smooth handling of important information like students, fees, and departments, making school operations more organized and faster.
- Streamlined Academic Tasks:
  By automating tasks such as registrations, and fee management, the system simplifies making them straight forward.

- User-Friendly Interface:
  With an easy-to-use design, the system ensures simplicity for all users. It doesn't require complicated learning, making it accessible and friendly for both administrators and end-users.
- Offline Accessibility:
  The system is designed to work without internet, catering to schools with limited connectivity. This not only ensures constant accessibility but also enhances data security for a reliable school management experience.

1.4 Scope and Limitations:

Scope:

This system is developed specifically for smaller colleges, focusing on simplifying student management tasks, such as adding student details and processing fee payments and identifying total students and fee to be collected.

Limitations:

Some of the limitations are:

- System isn’t multiuser.
- Data Loss may occur.
- Cannot generate receipt of payments.

UNIT 2: PROJECT DEVELOPMENT LIFE CYCLE

2.1 Introduction to Project development life cycle

Every project follows a structured development process, comprising distinct phases from inception to completion. These phases, collectively known as the Project Development Life Cycle While the standard project life cycle is widely recognized, professional services, which offer expertise to clients for profit, necessitate a more nuanced approach. This unique project cycle integrates recurring feedback loops to ensure each project's success, reflecting the dynamic nature of professional service engagements.

2.2 Project Planning and Feasibility Study:

Before investing in any software, it's essential to assess its suitability for our daily operations based on the required information. Project planning ensures that we chart a course that aligns with our needs. We prepare a feasibility report to present a thorough analysis of the project's viability and commercial potential.

2.3 Requirement Analysis

This is the one of the important phases of the PDLC. In this we come up with a detail report from the different fields for finding the business. We need to get the entire requirement regarding the problem otherwise it creates a lot problem in future. We need to be clear about the way to questions used to be recorded and how the users want those recorded in the system.

2.4 System Design

Design refers to the modules used to build in the software development. There are different modules such as E-R diagram, DFD’S, Flowchart etc. These modules help in the design of the program. According to the need of the programmer such modules are used.

2.4 Methodology

We have used Agile Model to develop this project. This model consists of following phases:

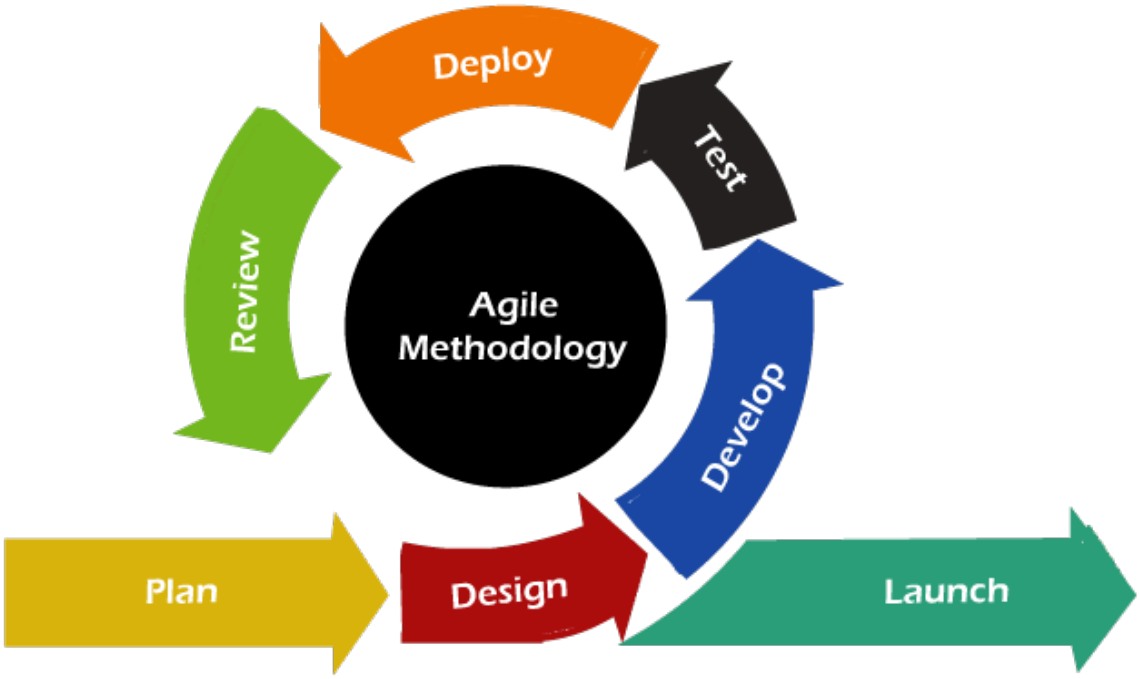


Fig 2.1 : Agile Model

2.5 Coding

On the basis of the system design, we did the coding of the system. The coding is done using the programming language. The technology used for coding in this project is VB.Net. It will develop in windows platform. Visual Basic is a general language and used as general purpose

2.6 Debugging

We have debugged our program for finding the errors and reducing the no. of bugs or defect in a computer or a piece of electronic hardware thus making it behave as expected. The basic steps that we have used in our program are:

- Isolate the source of the bug
- Identify the cause of bugs
- Determine a fix for the bugs
- Apply the fix and test it

2.7 Testing

Testing is the process of debugging of the software i.e., discovering the errors of the bugs and removing them. Actually, we have tested a program to work correctly, to discover the causes of these errors, and to revise the program code to eliminate the errors. Our tested program has a final measure of quality assurance for software product during the later phase of the system development life cycle. Thus, as we have tested our program using various techniques (testing processes), our software is free of errors and can serve the best.

2.8 Implementation

System implementation generally focuses on the coding and installing of the system. Our system implementation is composed of activities, which are coding, testing and installation. The purpose of these steps is to convert the physical system specification into the working and reliable software.

2.9 Documentation

We have all documents, all the activity performed during the development of the system, which will be very useful in the future modification and changes.

UNIT 3: SYSTEM REQUIREMENTS

3.1 Hardware Configuration

For smooth running of the project file the following minimum hardware is required.

- RAM: 4GB or above
- Hard Disk: 256GB or above
- Processor: i5 or above

3.2 Software Configuration

Choosing a programming language appropriate to the system is equally important as designing a system. There are many languages and packages available, and all have their own importance in their respective field. We develop our project in Windows environment by using the Visual Studio 2022. For smooth running of the project, the following minimum software is needed.

- Operating system: Windows 7/8/10
- Server: Microsoft SQL Server 2012 and later
- Framework: .NET Framework 4.6.1 and later

UNIT 4: TIME, COST AND TASK DIVISION

4.1: Cost Estimation and Duration:

Time, cost, and task division plays a vital role in the software development. So, the above factors should be properly considered while developing the software. The estimated time duration of this application is 1 month. We should develop such types of software that could be prepared within 1 month.

SN.	Activities	Price (Rs)
1	Data Collection	500
2	Internet Usage	1000
3	Transportation	200
4	Interface Design	300
5	Designing	500
6	Testing	300
7	Miscellaneous Expenses	500
	Total	3300

Table no 1.1: Cost Estimation

4.2: Task Division:

This project is prepared in a group. The name of the members of the project and their task division are given below:

SN.	Name of Students	Task Performed
1	Keshab Chidi, Nabin Chaudhary	Requirement analysis & finalizing
2	Ammar Bahadur Kunwar, Sudip Rana	Coding, and designing
3	Amrit Shrestha	Documentation
4	Nabin Chaudhary	Presentation

Table no 1.2: Task Division

UNIT 5: DEVELOPMENT TOOLS

5.1 Context Flow Diagram

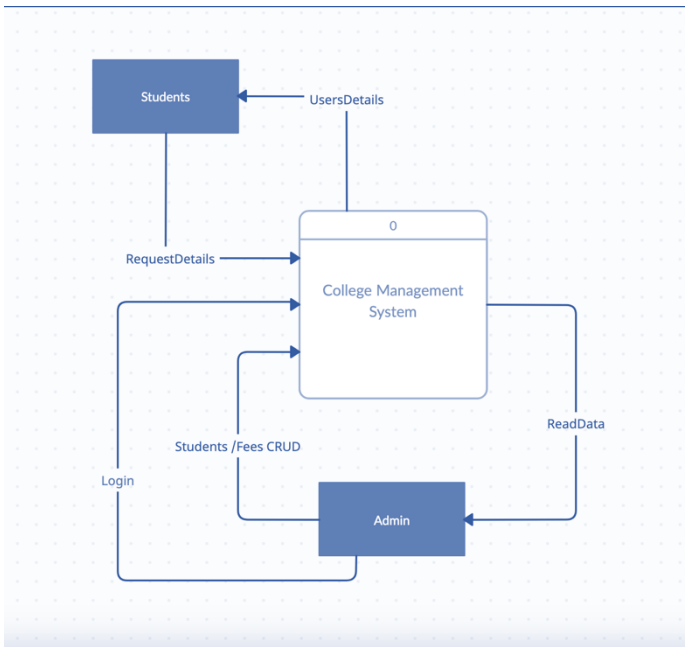


Fig 2.2: Context Flow Diagram

5.2 Data Flow Diagram

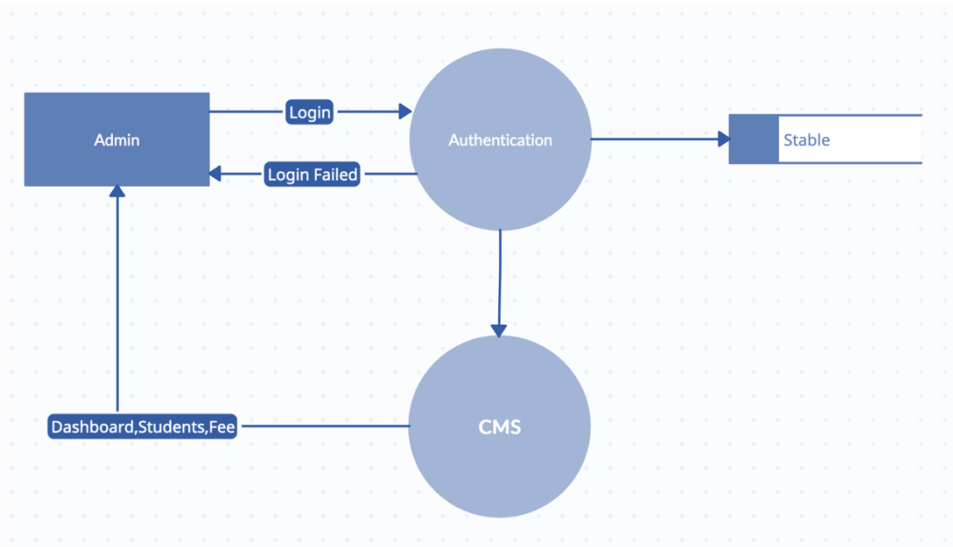


Fig 2.3: Level 1 DFD

5.2 ER Diagram

Fig 2.4: ER Diagram

5.3 Use Case Diagram

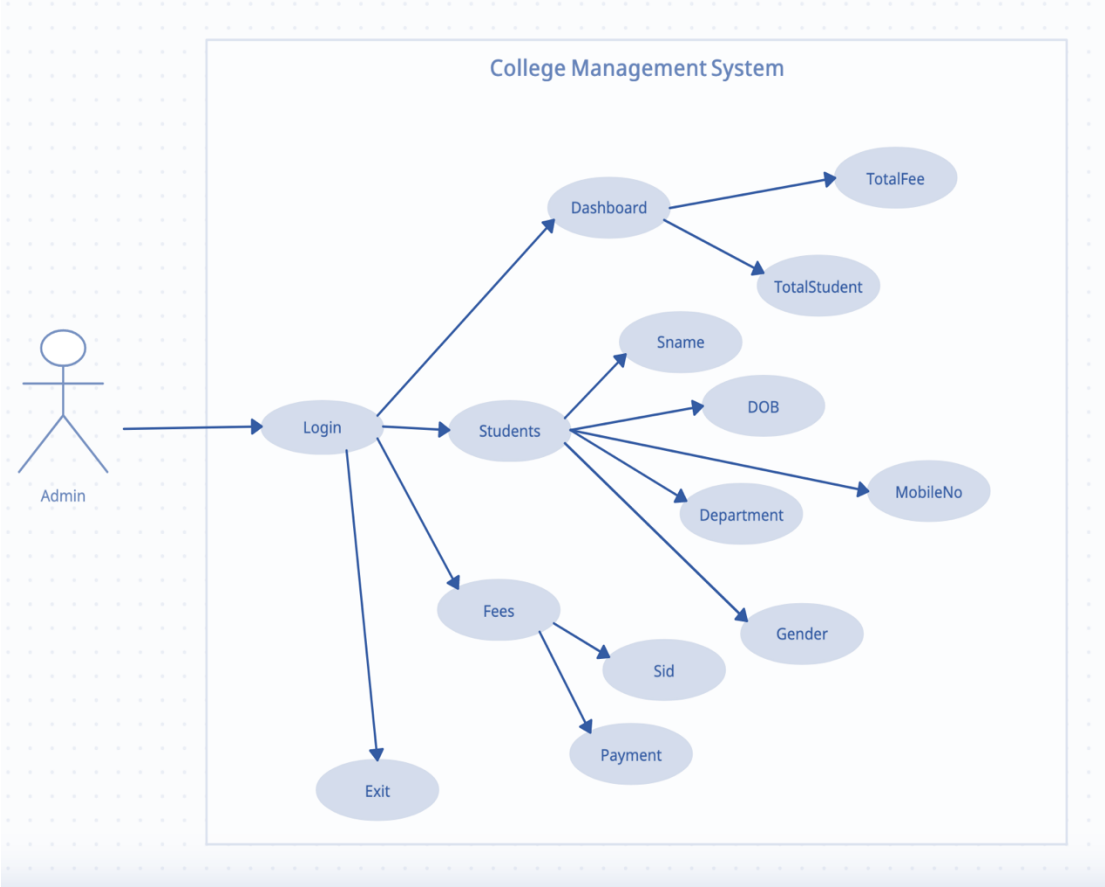


Fig 2.5: Use Case Diagram

UNIT 6: APPENDICES

6.1: Gantt Chart:

Tasks	Week 1	Week 2	Week 3	Week 4	Week 5
Project Planning &feasibility					
Requirement analysis					
System Design					
Coding					
Debugging					
Testing					
Implementation					
Documentation					

Fig 3.1: Gantt Chart

6.2: Snapshots:

Login Form

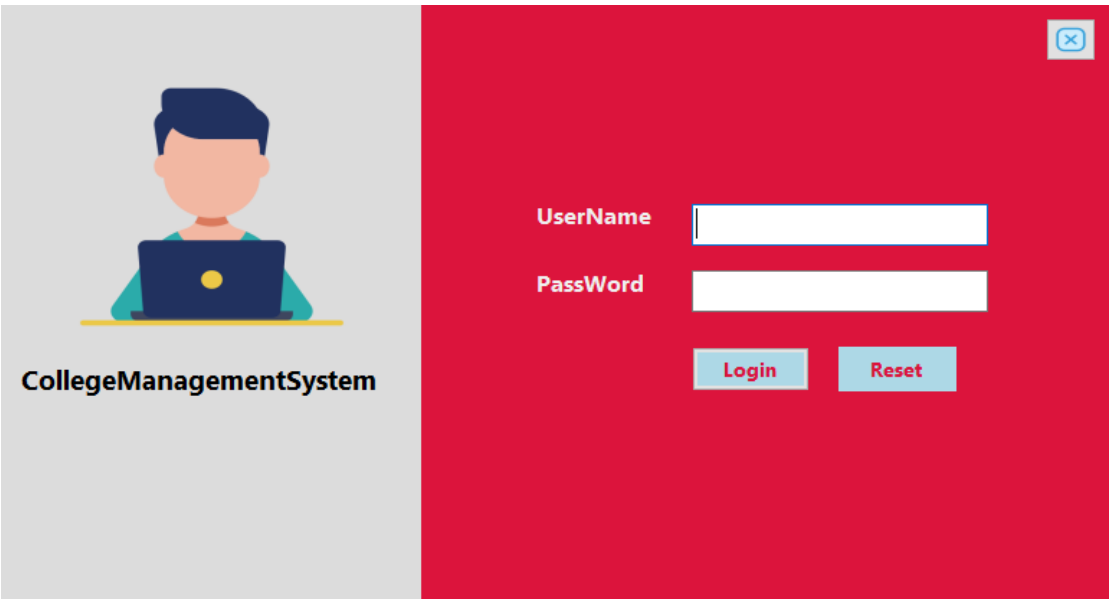


Fig 2.6: Login Form

Reset Form

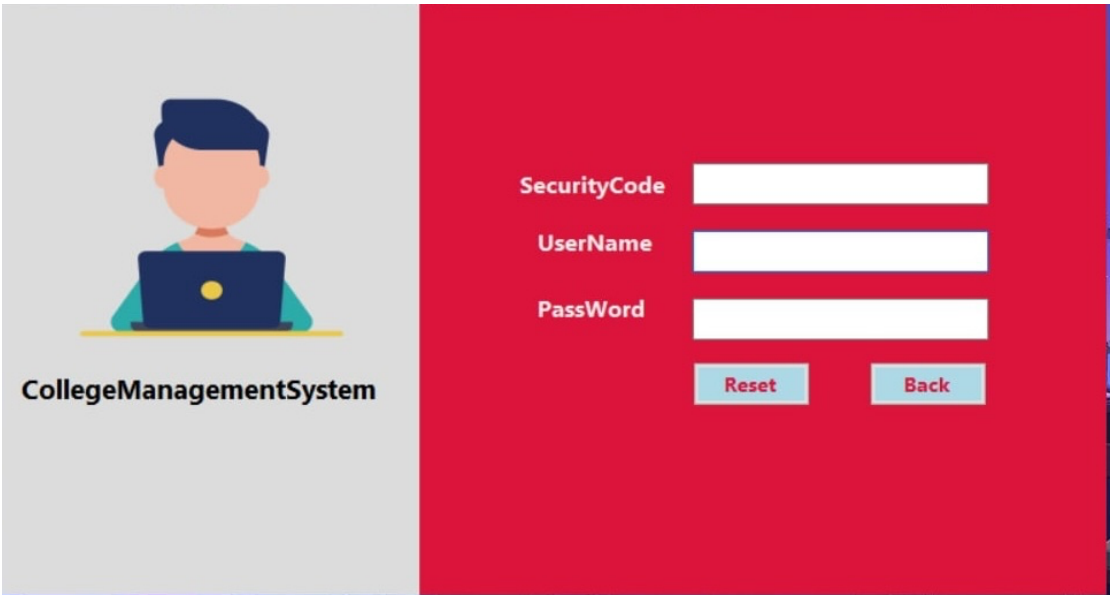


Fig 2.7: Reset Form

Dashboard Form

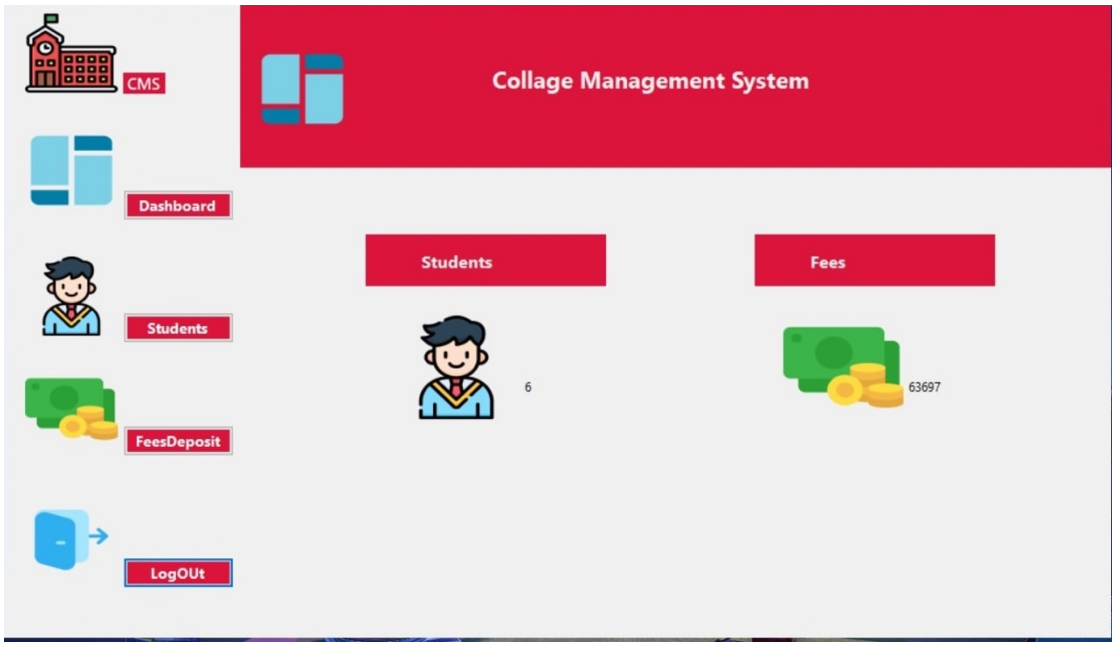
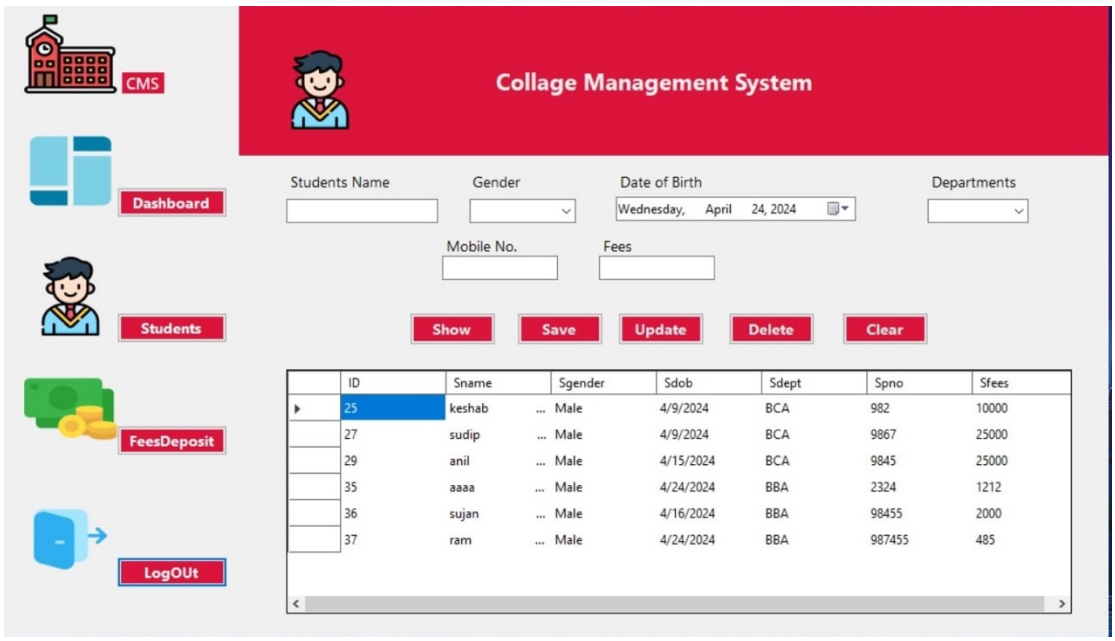


Fig 2.8: Dashboard Form

Student Form



ID	Sname	Sgender	Sdob	Sdept	Spno	Sfees
25	keshab	Male	4/9/2024	BCA	982	10000
27	sudip	Male	4/9/2024	BCA	9867	25000
29	anil	Male	4/15/2024	BCA	9845	25000
35	aaaa	Male	4/24/2024	BBA	2324	1212
36	sujan	Male	4/16/2024	BBA	98455	2000
37	ram	Male	4/24/2024	BBA	987455	485

Fig 2.9: Student Form

Payment Form

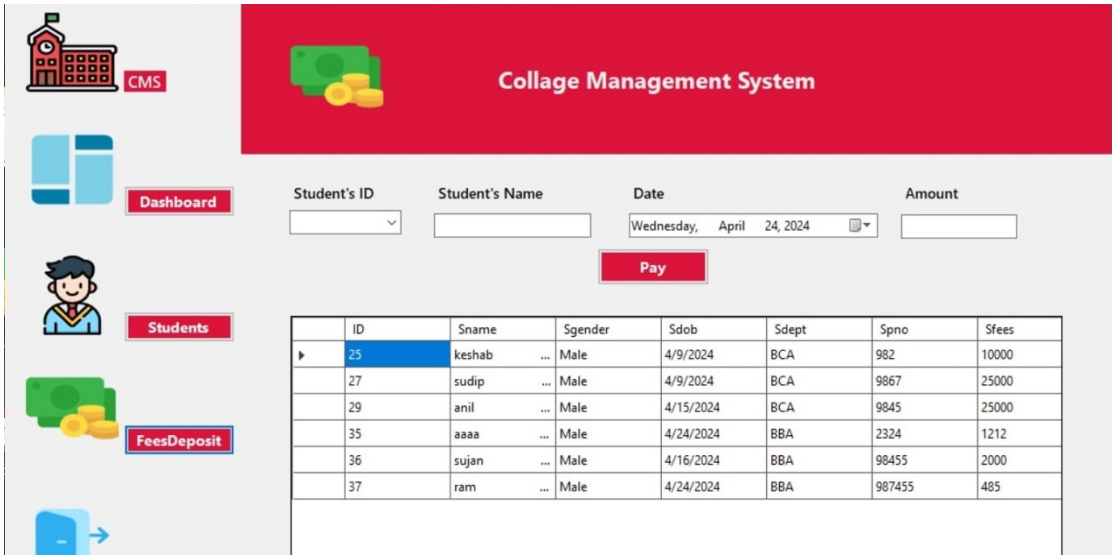


Fig 2.10: Payment Form

UNIT 7 CONCLUSION AND RECOMMENDATION

7.1 Conclusion

Building an awesome project is tough. It takes a lot of mind labor during each phase of Development cycle. Our project is built taking long time in designing and coding phase. We have taken a couple of sample projects during the designing and developing phase as well. It is built by using several modules and integrating them in a single piece. Our Project has achieved simplicity and hassle-free environment by the end of development phase. Using our system, store admin and users can easily deal with products and their issues.

Visual Basic is chosen as the front-end application. VB applications are typically compiled to make an executable file that can run on any computer regardless of computer architecture. Visual Basic is one of the most popular programming languages in use, particularly for client-server applications, with millions of users. SQL server is chosen for back-end application

7.2 Recommendations

Based on the afore mentioned description of this project, we recommend this software to be used in any mid and small colleges. Though it isn't perfect, it can prove to be quite good enough. It can easily solve the burden on manual data management by implementing the computerized form. For best results and performance of this software, we recommend all the necessary hardware and software requirements mentioned above to be fulfilled. If so, we are absolutely convinced that it won't let the operator down.

References / Bibliography

References

- https://docs.microsoft.com/en-us/
- https://stackoverflow.com/

Bibliography

- Steven Kogent,"VB.NET 2010 Programming Black Book", Dream tech pub.
- An Introduction to Programming Using Visual Basic, David I. Schneider, Pearson